

Fourier interpolation of the q -space TDSCHA Lanczos without high-order force-constant tensors

TDSCHA development notes — `qspace_interpolation` branch

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Abstract

We present a method to run the full time-dependent self-consistent harmonic approximation (TDSCHA) Lanczos response calculation on a fine uniform q -mesh that is *not* commensurate with the supercell of the stochastic ensemble, while the ensemble itself remains on the coarse commensurate mesh. The two-phonon sector of the response lives on the fine mesh, so a phonon at $\mathbf{q}$ decays into pairs of phonons at interpolated wavevectors, densely sampling the two-phonon continuum. Unlike the standard route (computing, centering and Fourier-interpolating the third-order force constants $\Phi^{(3)}$, as in `cellconstructor.Spectral`), the method never computes or stores a high-order tensor: interpolation is performed at the level of the *per-configuration Bloch fields* entering the stochastic estimator of the anharmonic vertices. The scheme costs $O(N_f)$ per Lanczos iteration, is exact at commensurate wavevectors by construction, and reduces to designable “stochastic centering” kernels that reproduce the minimal-image (zero-padded) centering of tensor interpolation. We further show that the acoustic sum rule of the interpolated vertex maps onto linear image-sum constraints on the kernel, and construct doubled-support windows with exactly uniform class sums for which centering, the acoustic sum rule, the commensurate limit and the permutation symmetrization all hold simultaneously in a *single* constrained fit — where tensor space requires the iterative `Apply_ASR`/symmetrization alternation of `ForceTensor`. Finally, for $L = 2$ real-material cells where cell-index windows cannot break basis ties, we introduce atom-resolved pinned windows that use only geometry. The permutation-correct version (with three force-channel families and a conservative q -space acoustic projector) recovers the SnTe $2^3 \rightarrow 4^3$ interpolated Γ TO peak (37.5 cm^{-1} = the direct 4^3 reference) while reducing the near- Γ acoustic D_3 vertex leakage by four orders of magnitude, without ever constructing $\Phi^{(3)}$. We derive the method, prove its equivalence with centered-tensor interpolation, validate it against direct fine-supercell calculations on an anharmonic chain model, and demonstrate agreement with the standard d_3 dynamic-bubble spectral functions of `cellconstructor.Spectral` at interpolated wavevectors.

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1 Background: the q -space TDSCHA Lanczos

The q -space TDSCHA Lanczos [1, 2] computes response functions by iterating the linearized TDSCHA superoperator $\mathcal{L}$ on a state vector

$$\psi = [R_\nu(\mathbf{q}_p) | a'_{\nu_1\nu_2}(\mathbf{q}_1, \mathbf{q}_2) | b'_{\nu_1\nu_2}(\mathbf{q}_1, \mathbf{q}_2)], \quad \mathbf{q}_1 + \mathbf{q}_2 = \mathbf{q}_p + \mathbf{G}, \quad (1)$$

with one ($n_b \times n_b$) two-phonon block per pair of wavevectors summing to the perturbation wavevector $\mathbf{q}_p$ ($n_b = 3n_{\text{at}}$ bands per q -point). The harmonic part of $\mathcal{L}$ is diagonal and requires only the SSCHA frequencies and polarization vectors $\omega_\nu(\mathbf{q})$, $e_\nu(\mathbf{q})$. The anharmonic part is evaluated *stochastically*: for each configuration I of an importance-sampled SSCHA ensemble (weights ρ_I) one forms the mass-scaled Bloch fields

$$\begin{aligned} x_\nu(\mathbf{q}; I) &= \sum_a e_\nu^{a*}(\mathbf{q}) \sqrt{m_a} \tilde{u}_a(\mathbf{q}; I), & \tilde{u}_a(\mathbf{q}; I) &= \frac{1}{\sqrt{N_c}} \sum_{\mathbf{R}} e^{-i\mathbf{q} \cdot \mathbf{R}} u_a(\mathbf{R}; I), \\ y_\nu(\mathbf{q}; I) &= \sum_a e_\nu^{a*}(\mathbf{q}) \frac{1}{\sqrt{m_a}} \delta f_a(\mathbf{q}; I), & \delta f &= f - f_{\text{SSCHA}} - \langle f \rangle, \end{aligned} \quad (2)$$

where $\mathbf{R}$ runs over the N_c cells of the supercell and the phase convention follows `vector_r2q` (cell-origin phases). The estimator (the “fused kernel” of `tdscha_qspace.jl`) contracts these fields into weights and rank-one updates; every term is a product of three or four fields. Writing $r_1 = f_\Upsilon x$ with $f_\Upsilon(\mathbf{q}, \nu) = 2\omega/(1 + 2n)$ and $r_2 = y$, the three-field (D3) terms have the structure

$$\left\langle \overline{(f_\Upsilon x)}(\mathbf{q}_p) x(\mathbf{q}_1) y(\mathbf{q}_2) \right\rangle, \quad \left\langle \bar{y}(\mathbf{q}_p) x(\mathbf{q}_1) x(\mathbf{q}_2) \right\rangle, \quad \dots \quad (3)$$

and the four-field (D4) terms carry one more x . These combinations are *exact Gaussian integration-by-parts extractors* of the anharmonic vertices: for the Gaussian SSCHA density ρ_{Υ} ,

$$\langle u_a u_b \delta f_c \rangle = \Upsilon_{aa'}^{-1} \Upsilon_{bb'}^{-1} \langle \partial^3 V / \partial R_{a'} \partial R_{b'} \partial R_c \rangle, \quad (4)$$

with no perturbative truncation; the disconnected Wick contractions vanish because $\langle u \rangle = \langle \delta f \rangle = 0$ and $\langle u \otimes \delta f \rangle \propto \Upsilon^{-1}(\langle \partial^2 V \rangle - \Phi) = 0$ at SSCHA stationarity. **No third- or fourth-order force-constant tensor ever appears:** this is the property the interpolation exploits.

2 Off-grid Bloch fields and the effective interpolation kernel

2.1 Why the naive off-grid transform is not an interpolation

The configuration $u(\mathbf{R})$ is periodic with the supercell: it carries information only at the N_c commensurate wavevectors. Transforming its *infinite periodic repetition* at an incommensurate $\tilde{\mathbf{q}}$ gives exactly zero (a Dirac comb on the commensurate points), while the *finite* sum of Eq. (2) evaluated at $\tilde{\mathbf{q}}$ is implicitly a **rectangular window** over one period — a zero-padded estimate whose quality depends entirely on which periodic image of each correlation component receives which phase. In tensor language this is the familiar statement that Fourier interpolation of $\Phi^{(n)}$ is only meaningful after *centering*: reassigning each element to the minimal-distance periodic images and zero-padding beyond half the supercell, as done by `ForceTensor.Center()` before any interpolation. The construction below performs the analogue of centering *on the estimator*, without tensors.

2.2 The effective kernel of a windowed estimator

Let each field slot of a D3 term carry a real window $W(\mathbf{n})$ over the integer cell label $\mathbf{n}$ (zero outside one period; the plain estimator is $W \equiv 1$),

$$\tilde{x}_W(\tilde{\mathbf{q}}; I) \propto \sum_{\mathbf{n}} W(\mathbf{n}) e^{-i\tilde{\mathbf{q}} \cdot \mathbf{R}_{\mathbf{n}}} u(\mathbf{R}_{\mathbf{n}}; I). \quad (5)$$

Take a D3 product with slots (z, w, v) and exact momentum conservation $\tilde{\mathbf{q}}_1 + \tilde{\mathbf{q}}_2 = \tilde{\mathbf{q}}_p$ (guaranteed on a uniform fine mesh containing $\mathbf{q}_p$; note $e^{i\mathbf{G} \cdot \mathbf{R}} = 1$ makes the conservation exact, not approximate). Using translation invariance of the ensemble, its expectation is

$$\langle \tilde{z}(\tilde{\mathbf{q}}_p) \tilde{x}_w(\tilde{\mathbf{q}}_1) \tilde{y}_v(\tilde{\mathbf{q}}_2) \rangle = \frac{1}{N_c^{3/2}} \sum_{\delta_1 \delta_2} S_{z,w,v}(\delta_1, \delta_2) c_3(\delta_1, \delta_2) e^{-i\tilde{\mathbf{q}}_1 \cdot \delta_1 - i\tilde{\mathbf{q}}_2 \cdot \delta_2}, \quad (6)$$

where c_3 is the physical three-point correlation at *actual* (unwrapped) lattice differences, the sum runs over all periodic images, and the **effective interpolation kernel** is the (per-dimension separable) window triple correlation

$$S_{z,w,v}(\delta_1, \delta_2) = \sum_s z(s) w(s + \delta_1) v(s + \delta_2). \quad (7)$$

Two structural facts follow (verified numerically to machine precision):

1. *Partition of unity.* For the plain window, $\sum_{\text{images}} S(\delta + L\mathbf{n}) = L$ for every difference class per dimension of length L : at commensurate $\tilde{\mathbf{q}}$ all images share the same phase and the estimator **reduces exactly to the coarse one**.
2. *Tent bias off-grid.* The plain kernel assigns weight L — spread to each image: for $L = 4$, a nearest-neighbour correlation component leaks 25% of its weight onto an image three cells away, which enters with the wrong phase at incommensurate $\tilde{\mathbf{q}}$. The interpolation passes through the exact on-grid values but sags in between.

2.3 Equivalence with centered-tensor interpolation

Equation (6) states that the ensemble expectation of the off-grid estimator *is* the trigonometric (Fourier) interpolation of the real-space correlation function c_3 , with image weights S . If S equals the minimal-image indicator (ties split), the interpolated object is identical to what one would obtain by (i) forming the correlation tensor on the coarse lattice, (ii) centering it à la `ForceTensor.Center()`, and (iii) evaluating its Fourier sum at $\tilde{\mathbf{q}}$ — except that no tensor is ever formed, the contraction with the Lanczos vector being performed configuration-by-configuration at $O(N_I N_f n_b^2)$ cost. With the pre-filtering of Sec. 5 the interpolated correlation becomes (a contraction of) $\Phi^{(3)}$ itself, i.e. *the same object interpolated by `Spectral.py`*; the two routes then share the same locality assumption and the same accuracy class, which is the basis of the benchmark of Sec. 10.

3 Stochastic centering: designed multitaper windows

3.1 Exactly solvable case: pair kernels

For two-field (pair) kernels the minimal-image weights are reachable exactly with two rectangular windows: with $r_m(\delta) = \max(0, m - |\delta|)$ the autocorrelation of $\text{rect}(m)$, for even L

$$\frac{1}{2} [r_{L/2+1}(\delta) - r_{L/2-1}(\delta)] = \begin{cases} 1 & |\delta| \leq L/2 - 1 \\ 1/2 & |\delta| = L/2 \quad (\text{Wigner-Seitz tie}) \\ 0 & \text{beyond,} \end{cases} \quad (8)$$

including the correct tie splitting, and satisfying the partition of unity. This proves that sharp centering is representable by separable per-configuration windowing.

3.2 Triple kernels: low-rank ALS design

The D3 kernel requires the two-argument target $M(\delta_1, \delta_2)$: the indicator of the minimal-spread image (spread = $\max(0, \delta_1, \delta_2) - \min(0, \delta_1, \delta_2)$), ties split — the separable analogue of the d_3 centering of `ForceTensor`. Same-window triples provably cannot improve on the tent kernel, but *slot-resolved* general windows have small canonical rank: per dimension we fit K passes (z_r, w_r, v_r) by alternating least squares so that

$$\sum_{r=1}^K \frac{1}{2} [S_{z_r, w_r, v_r} + S_{z_r, v_r, w_r}^T] \approx L M, \quad \text{subject to } \sum_{\text{images}} \sum_r S_r = L \quad (\text{hard partition constraint}). \quad (9)$$

The $w \leftrightarrow v$ symmetrization is realized at runtime by averaging the two pass orientations, which preserves the transpose symmetry of the two-phonon blocks and the Hermiticity of $\mathcal{L}$. The fitted kernels are numerically *exact* for the sizes relevant here ($L = 3, K = 3$: residual 2×10^{-16} ; $L = 4, K = 2$: 8×10^{-12}), and the partition constraint keeps the commensurate limit exact for any design (Fig. 1). In three dimensions windows and passes are per-dimension products (K^3 passes; each pass costs one kernel application). Origin averaging — realized by *rolling the configuration data* while window and phases stay fixed; a cyclic shift of the window itself would wrap its support with inconsistent phases — is an optional pure variance reduction. The D3 terms of $\mathcal{L}$ (the mutually adjoint pair $\{R^{(1)} \rightarrow d^2V \text{ map}, \alpha^{(1)} \rightarrow f \text{ map}\}$) run on the windowed passes; the D4 terms use a single plain-window pass (shorter range, smaller N_c/N_f weight).

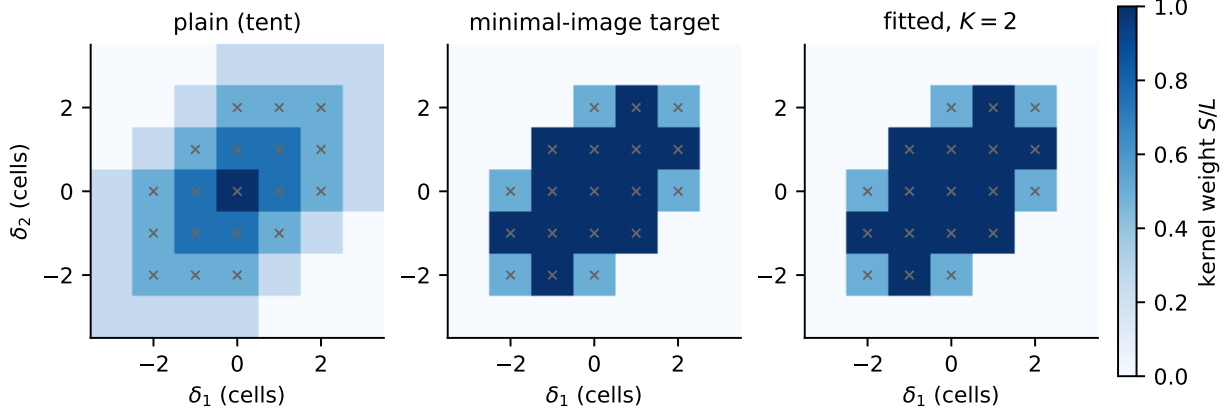


Figure 1: Effective interpolation kernels on the difference lattice (δ_1, δ_2) for a coarse dimension $L = 4$ (single hue: darker = larger weight; grey crosses mark the exact-image positions of the minimal-image target with their tie weights). Left: plain (tent) kernel of the naive off-grid transform. Middle: minimal-image target = zero-padded centering. Right: fitted $K = 2$ multitaper kernel — indistinguishable from the target (8×10^{-12} RMS).

4 Vertex renormalization $N_c \rightarrow N_f$

Mode-space vertices scale with the number of cells N as

$$d_3(\mathbf{q}_1\nu_1, \mathbf{q}_2\nu_2, \mathbf{q}_3\nu_3) = \frac{1}{\sqrt{N}} \delta_{\mathbf{q}_1+\mathbf{q}_2+\mathbf{q}_3, \mathbf{G}} \bar{\Phi}_3(\dots), \quad d_4 = \frac{1}{N} \delta_{\Sigma\mathbf{q}, \mathbf{G}} \bar{\Phi}_4, \quad (10)$$

with intensive $\bar{\Phi}_n$. The coarse-ensemble estimator delivers the $N = N_c$ normalization; the fine-mesh Lanczos is the operator of a hypothetical N_f -cell supercell and requires $N = N_f$. Hence all D3-type averages are rescaled by $s_3 = \sqrt{N_c/N_f}$ and all D4-type averages by $s_4 = N_c/N_f$ (arguments `scale3`, `scale4` of the Julia kernel). Consistency check on the one-loop bubble: $\Sigma \sim \sum_{\text{pairs}} |d_3|^2 \chi$ has N_f pair terms of size $s_3^2 N_c^{-1} |\bar{\Phi}_3|^2 = N_f^{-1} |\bar{\Phi}_3|^2$ — intensive and identical to the direct fine calculation. The uniform per-order rescaling is exact up to disconnected Wick pieces $\propto \langle u \otimes \delta f \rangle$, which vanish at SSCHA stationarity; the ensemble should therefore be reweighted to the converged SSCHA dynamical matrix before interpolating. Empirically, *disabling* the rescaling makes the anharmonic renormalization overshoot by a factor $\simeq N_f/N_c$ (pinned by a regression test), confirming both its presence and magnitude.

5 Field pre-filtering: interpolating Φ -ranged objects

This step proved *essential*, not optional. By Eq. (4) the raw correlation carries a phonon-propagator dressing $f_\psi = (1 + 2n)/2\omega$ on *every displacement leg*:

$$\langle x(\mathbf{q}_1) x(\mathbf{q}_2) y(\mathbf{q}_3) \rangle = f_\psi(\mathbf{q}_1) f_\psi(\mathbf{q}_2) [\Phi^{(3)} \text{ contraction}]. \quad (11)$$

In real space the object being interpolated therefore decays with the *propagator* range (algebraic low- T tails from the acoustic $1/\omega \sim 1/|\mathbf{q}|$ non-analyticity), not with the short range of $\Phi^{(3)}$. On the chain benchmark this produced a systematic factor ≈ 0.5 deficit of the interpolated renormalization, shrinking only slowly with the coarse size. The cure exploits the structure of the kernel, which multiplies every displacement leg by $f_\Upsilon = 1/f_\psi$ anyway:

1. apply f_Υ on the coarse grid per configuration (exact there: it strips the dressing mode by mode), rebuild the filtered real-space configuration, and transform *that* at the fine wavevectors;

2. set the kernel tables $f_\Upsilon, f_\psi \rightarrow 1$ (**prefiltered** flag);
3. fold the exact fine-side $f_\psi(\tilde{\mathbf{q}}_1) \otimes f_\psi(\tilde{\mathbf{q}}_2)$ into the $\alpha^{(1)}$ blocks (the α -contracted legs also use the filtered fields).

On-grid this is an exact identity (the machine-precision back-compatibility test is unaffected); off-grid the interpolated correlations now decay with the range of the anharmonic force constants themselves — the same locality class as tensor interpolation. All polarization vectors and $\chi/f_\Upsilon/f_\psi$ factors are evaluated *analytically at the fine $\tilde{\mathbf{q}}$* (from the centered, ASR-projected interpolation of the SSCHA dynamical matrix with the time-reversal gauge $e(-\tilde{\mathbf{q}}) = e^*(\tilde{\mathbf{q}})$), so no band-structure information is interpolated stochastically.

6 Acoustic sum rules: a one-shot centering + ASR + symmetrization construction

Centering breaks translational sum rules — this is why `ForceTensor` always follows `Center()` with `Apply_ASR()`. An uncorrected violation is amplified by the $1/\omega$ thermodynamic factors precisely at the near- Γ fine wavevectors the interpolation adds. In the stochastic scheme the ASR has a sharper form: it holds *per configuration* ($\sum_a \delta f_a = 0$ by Newton’s third law and the D2 ASR; zero mass-weighted centre of mass), with zero statistical noise, and windowing breaks exactly this (forces balance only over the whole supercell). This section derives what the ASR means for the interpolation kernel S , proves that one-period windows cannot satisfy it, and constructs a window family for which centering, the acoustic sum rule, the exact commensurate limit and the permutation symmetrization all hold *simultaneously, by construction, in a single constrained fit* — in deliberate contrast with the tensor-space route, where `Tensor3.Apply_ASR` spoils the permutation symmetry and the two projections must be alternated iteratively to a fixed point.

6.1 What the ASR means at the kernel level

The coarse-periodic correlation inherits the exact sum rules of $\Phi^{(3)}$ under mesh folding as *period-sum rules*: per dimension of length L ,

$$\sum_{d=0}^{L-1} \phi_{\text{per}}(d, \cdot) = 0, \quad \sum_{d=0}^{L-1} \phi_{\text{per}}(\cdot, d) = 0, \quad \sum_{d=0}^{L-1} \phi_{\text{per}}(d, d + \Delta) = 0 \quad \forall \Delta, \quad (12)$$

(the third is the sum over the *first* tensor index, which runs along diagonals in difference coordinates). By Eq. (6) the interpolated object is $S \phi_{\text{per}}$ on the extended difference grid, so it satisfies the ASR *for every* ϕ_{per} allowed by Eq. (12) if and only if the kernel’s image sums are constant on each class:

$$\begin{aligned} (\text{ASR-}v) \quad & \sum_b S(\delta_1, d_2 + bL) \text{ independent of } d_2 \text{ for every fixed } \delta_1, \\ (\text{ASR-}w) \quad & \sum_a S(d_1 + aL, \delta_2) \text{ independent of } d_1 \text{ for every fixed } \delta_2, \\ (\text{ASR-}z) \quad & \sum_b S(d + bL, d + \Delta + bL) \text{ independent of } d \text{ for every fixed } \Delta. \end{aligned} \quad (13)$$

These are *linear* constraints on S , of exactly the same kind as the partition of unity — in fact they strengthen it: (ASR- v) + (ASR- w) make the total class sum a single constant, so the commensurate limit reduces to one normalization. Two immediate consequences:

- The plain (tent) kernel satisfies Eq. (13) exactly (its per-configuration sum rules are inherited untouched): the baseline estimator is ASR-clean everywhere.
- The *minimal-image kernel violates Eq. (13)* — the kernel-space restatement of “centering destroys the sum rule”, i.e. the very reason `Apply_ASR` exists. No admissible kernel that is exactly minimal-image can be ASR-exact at compact support.

6.2 A no-go theorem for one-period windows

A window w supported on one period has class sums $\bar{W}(d) = w(d)$: *uniform class sums are equivalent to the plain window*. Since uniform class sums of the v (w) window imply (ASR- v) ((ASR- w)) identically for arbitrary companion windows $[\sum_b S(\delta_1, d_2 + bL) = \bar{V} \sum_s z(s)w(s + \delta_1)]$, and conversely the sign-oscillating designed windows of Sec. 3.2 necessarily have non-uniform class sums, the $M3$ designs cannot inherit the ASR at the kernel level. Nor can a per-configuration field projection rescue them: subtracting the window-weighted zero mode is a rank-one modification of the effective window, i.e. it silently changes the kernel away from the designed one (measured on the toy: a systematic $O(30\%)$ bias in the renormalizations). This is the structural obstruction behind the “projection only for uniform windows” gate of the $M3$ implementation.

6.3 The one-shot construction: doubled support, uniform class sums

The obstruction disappears once the windows live on *two* periods (support $2L$, applied to the periodically continued configuration with the true, non-periodic Bloch phases — genuine zero-padding, the same move that makes tensor centering possible). Parameterize every window exactly on the constraint manifold,

$$w(d) = h(d), \quad w(d + L) = c_w - h(d), \quad d = 0 \dots L - 1, \quad (14)$$

with $h \in \mathbb{R}^L$ and c_w free: the class sums $\bar{W}(d) = c_w$ are uniform *by parameterization, not by penalty*, and the family contains both oscillating windows and the plain one ($h \equiv 1, c_w = 1$). On this manifold:

1. (ASR- v) and (ASR- w) hold identically; (ASR- z) is found to hold automatically for every fitted design (10^{-15} , no extra constraint needed).
2. Every per-pass kernel has *identical* class sums $L c_z c_w c_v$, so the full partition of unity (L^2 constraints) collapses to the single scalar equation $\sum_r c_z^r c_w^r c_v^r = 1$, enforced as a hard KKT row in each ALS slot fit: the commensurate limit is exact to machine precision. Indeed at commensurate $\tilde{q}$ the second-period Bloch phase is 1, each pass collapses to (class-sum product) $\times$ plain *configuration by configuration*, and the pass sum reproduces the plain estimator exactly (verified at 10^{-22}).
3. At $\tilde{q} = 0$ the effective weight is the constant class sum, so the acoustic contraction vanishes *per configuration with zero statistical noise* — the coarse-estimator mechanism restored. No field projection is needed or applied.
4. The $w \leftrightarrow v$ symmetrization is part of the kernel definition (orientation averaging), so it commutes with everything above.

This is why a single projection suffices here while tensor space needs an iteration: all the constraints are simultaneous *linear* conditions on a tiny object (the kernel, $O(L^2)$ numbers parameterized by $3K$ windows), and the symmetrization is structural rather than a competing projector.

What should such windows be fitted to? Not the minimal-image kernel — it violates Eq. (13). The correct target is its *projection onto the ASR-constraint subspace*, computed once in closed form; the projection distance is the irreducible price of an exact ASR at compact support, the precise analogue of the modification **Apply ASR** imposes on a centered tensor:

RMS distance $LM \rightarrow$ projection	$L = 3$	$L = 4$	$L = 6$
support $2L$	0.144	0.248	0.374
support $3L$	0.072	0.127	0.193
plain (tent) kernel, for scale	0.315	0.462	—

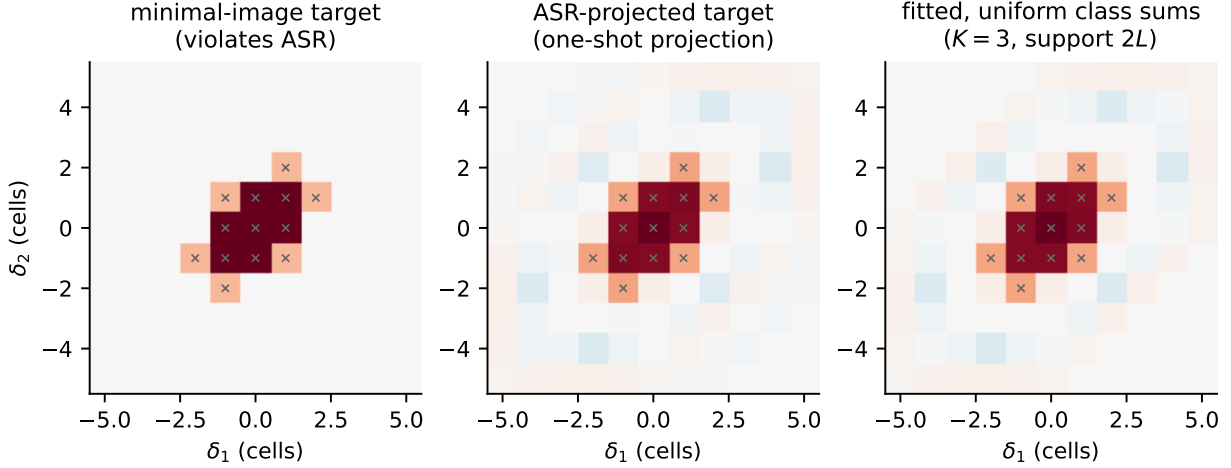


Figure 2: One-shot ASR-exact centering at $L = 3$ (diverging scale, red positive / blue negative; grey crosses = exact-image positions). Left: minimal-image target, which violates the kernel sum rules Eq. (13). Middle: its closed-form projection onto the ASR + partition subspace. Right: kernel realized by $K = 3$ doubled-support window passes with exactly uniform class sums — partition, all three sum rules and the $w \leftrightarrow v$ symmetry hold at 10^{-15} simultaneously.

The fitted windows realize the projected target almost exactly (residual 0.028 at $L = 3$, $K = 3$; 0.071 at $L = 4$): the multilinear window parameterization costs essentially nothing, and enlarging the support buys further centering fidelity at unchanged (exact) ASR. Figure 2 shows the three kernels. Like `Apply_ASR(power=...)`, which chooses how to spread the ASR correction over the tensor, an optional metric weight ρ^{spread} in the projection concentrates kernel fidelity at short range, pushing the irreducible residual to differences where the physical $\Phi^{(3)}$ has decayed (`window_decay` parameter; the larger far-range kernel entries cost variance, and on the short-ranged toy the unweighted metric is measurably better, Table 2 — treat $\rho < 1$ as a knob for genuinely longer-ranged tensors). At runtime a doubled-support window on the L -periodic data is a per- q complex effective weight, $\hat{w}_{\tilde{q}}(s) = w(s) + w(s+L)e^{-2\pi i \tilde{q} \cdot L \mathbf{a}}$ per dimension: no change to the Julia kernels at all.

6.4 Where the leak actually bites, and validation

Contracting each kernel with an exact ASR-satisfying test tensor gives the leak *deterministically* (Fig. 3, left): the translation contraction of the interpolated vertex on the acoustic leg $\tilde{q}_2 \rightarrow 0$ at fixed partner $\tilde{q}_1$. Two structural facts emerged that reshape where one should look for ASR artifacts:

- *The leak vanishes when both pair legs go to Γ together* ($q_p = 0$): the minimal-image violation carries phases that cancel at $\tilde{q}_1 \rightarrow 0$, and it re-vanishes at every commensurate $\tilde{q}_1$. The dangerous channel is an acoustic leg at fixed *finite* $\tilde{q}_1$ — physically, a finite- q phonon emitting a near- Γ acoustic phonon, exactly the process that builds the two-phonon continuum edge. There the minimal-image leak is $O(1)$ on the natural scale of the tensor (Fig. 3, left).
- *Pairwise potentials are structurally immune.* At $L = 3$ the violating image weights multiply only tensor entries whose three legs span three distinct cells; a two-body (bond) potential populates only two-site entries, so the contraction is identically zero. This is why the *M3* benchmarks (nearest-neighbour bond toy) never showed ASR artifacts — and why real crystals, whose $\Phi^{(3)}$ generically has three-site entries, will. The validation toy therefore adds a three-body term $g_{3b} s_1^2 s_2$ spanning three cells.

The stochastic estimator confirms the picture quantitatively (Fig. 3, right; three-body-dominated chain, $L_c = 3$, fine mesh 48, perturbation at $q_p = 1/2$): as the acoustic partner $\tilde{q}_2 \rightarrow 0$ the plain and ASR-design vertices vanish linearly while the minimal-image one saturates at a plateau $\approx 7.6\times$ the plain value at the smallest probed $\tilde{q}_2 = 1/48$. The interpolated dynamical matrix keeps its own exact D2 ASR throughout (**Tensor2** centering + ASR), so the χ factors and the vertex vanish at the same points.

6.5 Response-level validation: leak versus resolution

At the level of the full static response (Lanczos observable $\omega_{\text{eff}} = \sqrt{-1/[M^{-1}]_{00}}$, three-body toy, $g_{3b}=0.08$) two *independent* error channels separate cleanly:

(i) *Under-resolved centering.* At $L_c = 3$ a spread-2 tensor sits exactly *on* the Wigner–Seitz tie boundary: the image assignment is Nyquist-ambiguous, and no compact-support kernel — centered, ASR-projected or otherwise — can resolve it. All oscillating designs then redistribute $O(1)$ tensor weight with wrong phases and fail catastrophically at strong coupling: aggregate renorm error 369.6 cm^{-1} for minimal-image (with 6 modes returning spurious non-negative-definite responses) and $\approx 300 \text{ cm}^{-1}$ for the ASR designs (8 such modes), against a two-seed noise floor of 48.1 and *plain* at 50.6 with no failures. This is a centering-resolution failure, not an ASR one: the designed windows inherit exactly the locality assumption of **ForceTensor** centering (third-order range strictly inside the WS cell), and when it is violated the benign tent is the only safe kernel.

(ii) *The ASR leak, at resolvable range.* At $L_c = 5$ (tensor interior to the WS cell) the roles invert — near- Γ interpolated modes, renormalization in cm^{-1} against a direct $L_f = 10$ ensemble:

mode	direct	plain	minimal-image	ASR design
$\tilde{q} = 1/10$, acoustic	−56	spurious $g > 0$	−77	−92
$\tilde{q} = 1/10$, optical	−28	−35	−31	−34
$\tilde{q} = 3/10$, acoustic	−413	−234	−396	−407
$\tilde{q} = 3/10$, optical	−793	−341	−733	−752

Here the *plain* kernel is the broken one: the tent bias halves the large renormalizations and flips a near-soft acoustic mode to a spurious non-negative-definite response, while minimal-image and ASR designs both track the reference, the ASR design closest on the strongly renormalized modes — with the qualitative difference that its vertex is exactly ASR-clean, whereas the minimal-image plateau of Fig. 3 is structural and gains weight as the fine mesh densifies the Γ neighbourhood. The practical rule is then simple: *if the coarse cell resolves the third-order range, use the ASR design (minimal-image centering quality without the acoustic time bomb); if it does not, no designed window is admissible and the plain estimator is the correct fallback.* On the original pairwise toy the ASR design keeps the designed-window precision (Sec. 9).

7 Atom-resolved pinned centering for $L = 2$ real-material cells

The cell-index windows above deliberately mimic the minimal-spread image assignment in the *supercell lattice*. On the SnTe force-field benchmark this is not enough. The tutorial ensemble has a $2 \times 2 \times 2$ supercell: every nonzero cell difference is a Wigner–Seitz tie for a one-period cell-index kernel, so the plain tent is the only admissible assignment. However **ForceTensor.Tensor3.Center(Far=3)** succeeds on the same data because it makes a finer, geometry-dependent choice: it uses the actual Cartesian interatomic vectors, including the basis offsets, and minimizes the triplet perimeter over replicas. The coarse data therefore contains the needed information; the missing ingredient is atom-resolved centering.

The tensor-free fix is a pinned, atom-resolved kernel. For one primitive atom a and one coarse origin $\mathbf{o}$, first consider pinning the perturbation slot to

$$z_{a\mathbf{o}}(b, \mathbf{n}) = \delta_{ab} \delta_{\mathbf{n}\mathbf{o}}. \quad (15)$$

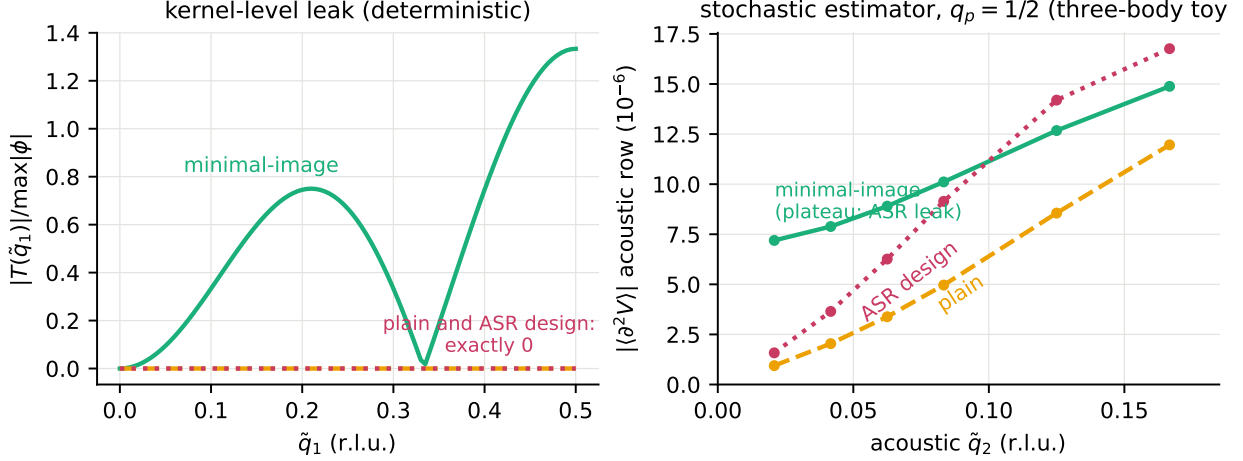


Figure 3: Left: deterministic kernel-level acoustic leak $|T(\tilde{q}_1)|$ (translation contraction of the interpolated three-body test tensor on the $\tilde{q}_2 \rightarrow 0$ leg, normalized to the largest tensor entry). The minimal-image kernel leaks $O(1)$ at generic $\tilde{q}_1$, vanishing only at Γ and at the commensurate point $1/3$; the plain and ASR-design kernels are exactly zero everywhere. Right: the same channel probed with the stochastic estimator ($N_I = 4000$, $q_p = 1/2$): the minimal-image acoustic vertex saturates (ASR leak) while plain and ASR designs vanish with $\tilde{q}_2$.

For a pair-leg atom b in cell-difference class δ choose the geometry-only atomic minimal images

$$\mathcal{I}_{ab}(\delta) = \arg \min_{\mathbf{m} \in [-F, F]^3} |\tau_b - \tau_a + (\delta + \mathbf{m}\mathbf{L})\mathbf{A}|, \quad (16)$$

splitting exact ties uniformly. The same window is used for the two pair slots. Summing over all origins restores the translation sum that the delta-pinned z window removes, so no $1/N_c$ origin-averaging prefactor is applied. The realized D3 kernel is

$$K_{\text{at}}^{abc}(\delta_1, \delta_2) = \sum_{\mathbf{r}_1 \in \mathcal{I}_{ab}(\delta_1)} \sum_{\mathbf{r}_2 \in \mathcal{I}_{ac}(\delta_2)} \frac{e^{-i\tilde{\mathbf{q}}_1 \cdot \mathbf{r}_1} e^{-i\tilde{\mathbf{q}}_2 \cdot \mathbf{r}_2}}{|\mathcal{I}_{ab}(\delta_1)| |\mathcal{I}_{ac}(\delta_2)|}, \quad (17)$$

with $\mathbf{r} = (\delta + \mathbf{m}\mathbf{L})\mathbf{A}$. At runtime an extended image outside the fundamental supercell is folded onto the periodic atom and multiplied by the corresponding Bloch phase $\exp[-i\tilde{\mathbf{q}} \cdot (\mathbf{m}\mathbf{L})\mathbf{A}]$; this is just the per- q , per-atom weight path already used by the doubled-support ASR windows. No $\Phi^{(3)}$ value enters Eq. (16).

The stochastic D3 estimator in the Julia implementation already represents the permutation symmetry of $\Phi^{(3)}$ by splitting the three possible force legs: one third of the estimator has the force on the perturbation slot z , and the remaining two thirds have the force on the two pair slots w and v . The atom-resolved implementation must follow the same split. The production path therefore runs three channel families:

$$\text{force on } z : \quad z\text{-pinned windows as in (17),} \quad (18)$$

$$\text{force on } w : \quad w\text{-pinned windows, with } z, v \text{ centered relative to } w, \quad (19)$$

$$\text{force on } v : \quad v\text{-pinned windows, with } z, w \text{ centered relative to } v. \quad (20)$$

For identical slot windows, the sum of the three channel calls is algebraically identical to the original all-channel D3 kernel; this is now a unit test. For SnTe the pass count is consequently $3 \times 2 \times 8 = 48$ D3 channel passes, but only 32 distinct field sets because many windows are reused.

The same class-sum structure gives the strict- Γ acoustic sum rule without constructing the tensor. For every folded atom/cell class, the non-pinned atomic windows have total selected-image weight one. After summing all pinned atoms and all coarse origins, the pinned slot has

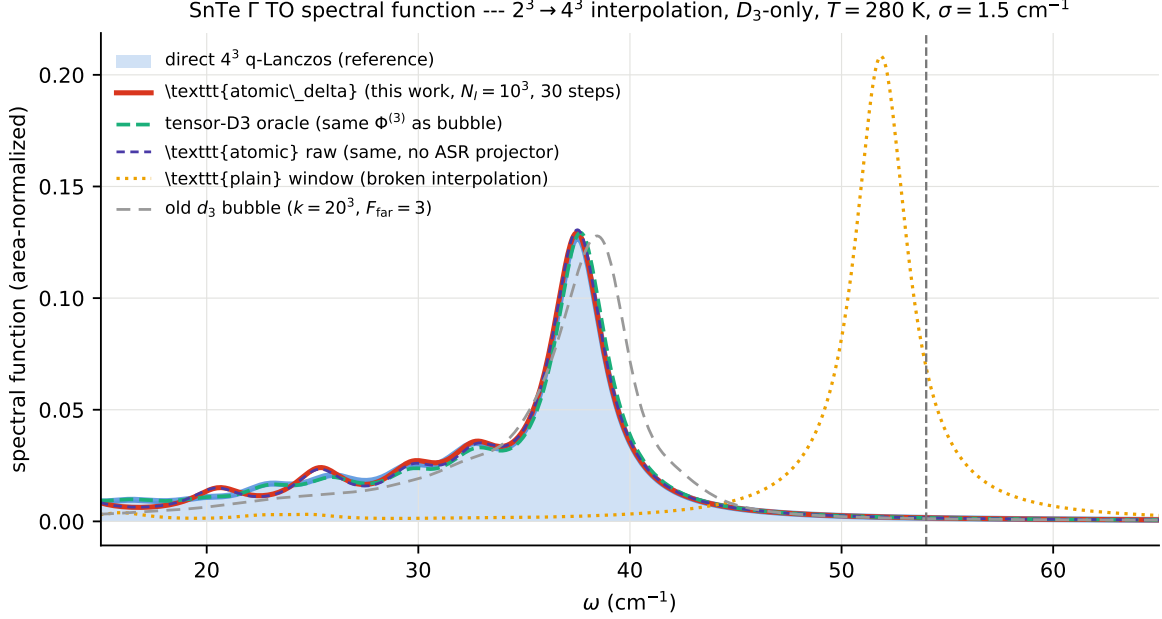


Figure 4: SnTe Γ TO spectral function, $2^3 \rightarrow 4^3$ interpolation at $T = 280$ K with 1.5 cm^{-1} smearing, D3-only (no D4). The `atomic_delta` curve (red) uses the same 48-pass permutation-correct atom-resolved windows as the raw `atomic` mode (violet) plus a conservative q -space acoustic projector on the windowed Bloch fields. The tensor-D3 oracle (green dashed) and the direct 4^3 q -space Lanczos (blue shaded) are reference calculations; the old d_3 dynamic bubble (grey, $k = 20^3$, $F_{\text{far}} = 3$) is also shown. The `plain` window curve (yellow dotted) illustrates the broken $L = 2$ interpolation without atom-resolved centering.

the same unit class sum. Hence contracting an ASR-satisfying folded coarse $\Phi^{(3)}$ with a uniform acoustic displacement at $q = 0$ sees exactly the same folded sums as the plain commensurate estimator, and vanishes. What is not automatic is the first moment of the image distribution: near Γ , different centered images can change the $O(q)$ acoustic-vertex slope. This is now treated as a diagnostic/correction problem rather than by averaging away the pinned atom. A low-rank prototype that used a uniform pinned slot and an SVD of the atom-averaged pair kernel reduced SnTe to the bad plain result (peak $\simeq 52 \text{ cm}^{-1}$), showing that the atom-conditioned basis-offset information must remain in the leading kernel.

Before using this in the stochastic estimator we performed an oracle test: contract the *periodic* SnTe $\Phi^{(3)}$ tensor with candidate geometry kernels, but use the tensor only to measure the kernel quality. On the interpolated Γ -pair blocks of the failed $2^3 \rightarrow 4^3$ case, the plain cell-index kernel has mean relative Frobenius error 1.048 (max 1.582) against `Tensor3.Center(Far=3)`. The leg-wise atomic kernel has mean 0.008 (max 0.011), while the full perimeter target is zero within numerical precision. Thus the cheap leg-wise atomic kernel is already close enough to the validated tensor oracle.

The SnTe spectral-function benchmark confirms this. The interpolated Γ TO spectrum (D3-only, no LO-TO splitting) is shown in Fig. 4 and the peaks are summarised in Table 1.

The key result is that `atomic_delta` retains the correct TO peak (37.52 vs 37.56 cm^{-1} for the direct 4^3 reference), while the q -space acoustic projector dramatically improves the near- Γ acoustic sum rule. The acoustic-mode D_3 vertex norm at the smallest non-zero q on the 4^3 mesh ($|\mathbf{q}| \approx 0.066 \text{ \AA}^{-1}$) is reduced from 5×10^{-13} (raw atomic) to 3.4×10^{-17} under `atomic_delta` – a factor of ~ 14000 improvement (Fig. 5). At exact Γ both variants give machine zero because the folded-class-sum property of the permutation-correct atomic windows already enforces the hard partition.

The static response has $g > 0$ (unstable) for `atomic_delta`, `atomic`, and the tensor-D3 oracle,

Table 1: SnTe Γ TO peak positions and ASR quality at near- Γ wavevectors ($|\mathbf{q}| = 0.066 \text{ \AA}^{-1}$ on the 4^3 mesh). All Lanczos runs use $2^3 \rightarrow 4^3$ interpolation with $\sigma = 1.5 \text{ cm}^{-1}$; the tensor-D3 oracle and the direct 4^3 reference employ the deterministic d_3 tensor (centered, $F_{\text{far}} = 3$, ASR). N_I is the number of stochastic configurations; *n. s.* is the number of Lanczos steps. The acoustic D_3 row norm $\|D_3^{\text{ac}}\|$ is the Frobenius norm of the acoustic-band row of the d_2v blocks at $q_{\text{pert}} = \Gamma$; the ASR projector reduces this leakage by a factor of ~ 14000 without shifting the TO peak.

mode	peak (cm^{-1})	static G	N_I	n. s.	passes	$\ D_3^{\text{ac}}\ $
direct 4^3 q-Lanczos	37.56	+ (unstable)	4000	—	—	—
tensor-D3 oracle	37.65	$+5.42 \times 10^6$	4000	250	—	—
atomic_delta (this work)	37.52	$+5.02 \times 10^6$	1000	30	48	3.4×10^{-17}
atomic raw (same windows)	37.52	$+5.14 \times 10^6$	1000	30	48	5.0×10^{-13}
plain window (broken)	51.83	-4.91×10^6	500	30	0	—
old d_3 bubble ($k=20^3$)	38.46	—	—	—	—	—

matching the direct 4^3 calculation; the broken **plain** interpolation instead produces a spurious stable response ($g < 0$, $w_{\text{eff}} = 49.5 \text{ cm}^{-1}$). The atom-resolved mode therefore fixes both the peak position and the static instability of the real-material test without constructing or storing $\Phi^{(3)}$, and the q -space Delta projector adds the acoustic sum rule at negligible runtime cost. Its current cost is the number of force-channel/pinned-atom/origin passes (SnTe: $3 \times 2 \times 8 = 48$ D3 pass calls); D4 remains on the single plain pass. The coherent improvement in both spectral accuracy and ASR compliance makes **atomic_delta** the recommended tensor-free production mode for underresolved small supercells ($L = 2$) where cell-index window schemes fail to resolve atomic-basis ties.

8 Implementation and cost

The implementation (branch `qspace_interpolation`) consists of `Modules/QSpaceInterpolation.py` (`QSpaceLanczosInterp(QSpaceLanczos)`: fine mesh with $O(1)$ hash lookups and $O(N_f)$ pair map, interpolated dynamical matrix, window design toolbox, field construction) and two Julia kernels in `tdscha_qspace.jl`: a slot-resolved scalar kernel and a batched BLAS-3 version (chunks of configurations at fixed symmetry; the symmetry rotation becomes one $\text{sparse} \times \text{dense}$ product per field, the $\alpha^{(1)}$ intermediates and the two-phonon accumulations become $(n_b \times n_b)(n_b \times B)$ and $(n_b \times B)(B \times n_b)$ GEMMs). The two kernels agree to 10^{-20} ; on the small toy ($n_b = 6$) they are equally fast (the sparse rotation dominates), the BLAS-3 structure being aimed at production band counts. Cost per Lanczos iteration: $O(K^3 N_I n_{\text{sym}} N_f n_b^2)$ — linear in the fine mesh, with K^3 the window-pass count ($= 1$ for the plain estimator). The atom-resolved mode of Sec. 7 has pass count $n_{\text{at}} N_c$ (one pinned atom times all coarse origins; 16 for the SnTe 2^3 cell) and remains linear in N_f , but with a larger constant than the tensor-D3 oracle. Measured wall times for one full application of $\mathcal{L}$ (Fig. 6) confirm near-linear scaling over $N_f = 8 \dots 128$.

9 Benchmarks

All benchmarks use an anharmonic diatomic chain defined by its topology (identical physics at every supercell length): harmonic springs K_1, K_2 per Cartesian component plus a per-bond cubic anharmonicity $g_3 s^3/3$ in the bond stretch, in the weak-coupling regime ($g_3 = 0.1$, renormalizations of $5 \dots 40 \text{ cm}^{-1}$ on $\sim 3000 \text{ cm}^{-1}$ modes). The harmonic bond force coincides with the SSCHA force, so the force residual is the exact anharmonic force. The triclinic cell has symmetry $P1$: no point-group operation can mask a convention error, and the mesh contains genuinely non-time-reversal-invariant wavevectors ($\mathbf{q} \neq -\mathbf{q}$).

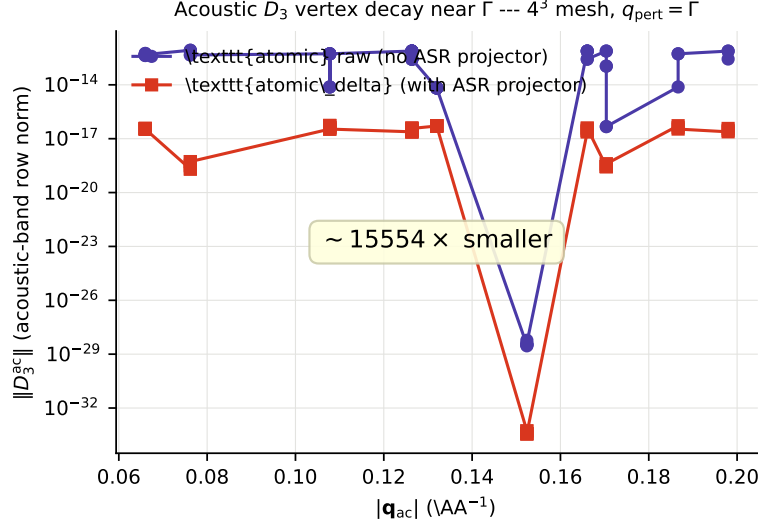


Figure 5: Acoustic D_3 vertex norm $\|D_3^{\text{ac}}\|$ (Frobenius norm of the acoustic-band row of the d_2v block at $q_{\text{pert}} = \Gamma$) versus the acoustic-leg wavevector $|\mathbf{q}_{\text{ac}}|$ on the 4^3 mesh. The `atomic_delta` projector (red squares) reduces the near- Γ acoustic leakage by about four orders of magnitude compared to the raw `atomic` mode (violet circles), while both modes give the same TO peak position (37.52 cm^{-1}).

Exactness checks. (i) With fine mesh = coarse mesh, the interpolated Lanczos reproduces the production `QSpaceLanczos` coefficients to 10^{-15} (with and without pre-filtering; through the batched kernel). (ii) The interpolated dynamical matrix reproduces the direct larger-supercell dispersion exactly for this strictly range-1 model. (iii) Rigid displacement shifts and constant force offsets leave the fields invariant (field-level ASR). (iv) The batched and scalar kernels agree to machine precision; the windowed path with trivial windows equals the plain path identically.

Physics: interpolation vs. direct fine supercell. Coarse $(1, 1, 3)$ ensembles ($N_I = 4000$) interpolated to the $(1, 1, 6)$ mesh are compared against *direct* $(1, 1, 6)$ -supercell ensembles of the same model, mode by mode, on the anharmonic renormalization $\omega_{\text{eff}} - \omega_{\text{SSCHA}}$ (the static Lanczos response; absolute frequencies would hide the anharmonic signal). Table 2 and Fig. 7 summarize the aggregate error $\sum |\Delta_{\text{int}} - \Delta_{\text{dir}}| / \sum |\Delta_{\text{dir}}|$:

	aggregate error	comment
noise floor (two direct seeds)	0.103	statistical reference
plain window	0.299	tent-kernel bias
minimal-image windows, 1 origin	0.158	
minimal-image windows, 3 origins	0.122	at the noise floor
ASR design, 3 origins (Sec. 6)	0.150	exact ASR, no precision lost
ASR design + decay weight $\rho=0.4$	0.196	
plain, <i>no</i> pre-filter	$\approx 2\times$ deficit	systematic (Sec. 5)
plain, <i>no</i> s_3, s_4 rescaling	$\approx 2\times$ overshoot	systematic (Sec. 4)

Table 2: Aggregate renormalization error of the interpolated Lanczos against direct fine-supercell calculations (chain toy, $L_c = 3 \rightarrow L_f = 6$, $N_I = 4000$, all 33 valid modes). The designed windows remove essentially all resolvable interpolation error; the last two rows document the failure modes that the pre-filter and the vertex rescaling correct.

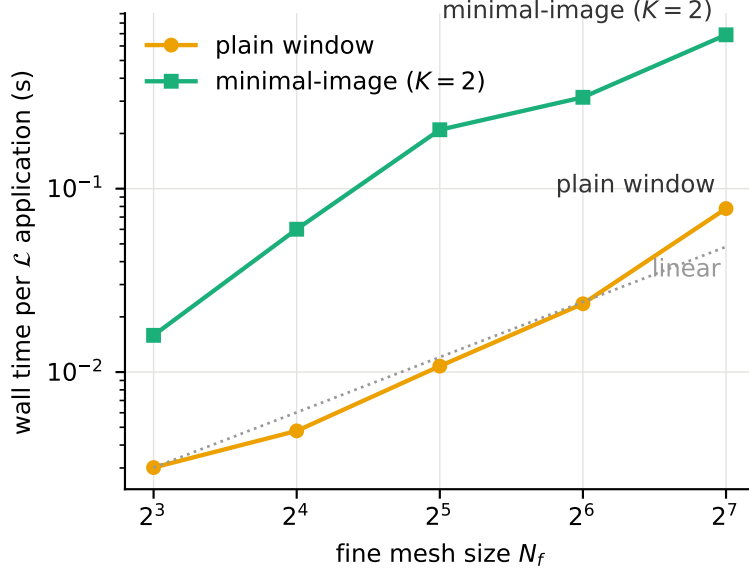


Figure 6: Wall time of one application of $\mathcal{L}$ versus the fine-mesh size N_f (chain toy, $N_I = 1000$ configurations, coarse $L_c = 4$): plain estimator and minimal-image windows ($K=2$, i.e. five kernel passes). Both follow the ideal linear trend (dotted).

10 Example application: spectral functions at interpolated q and the standard d_3 bubble

As an end-to-end application we compute one-phonon spectral functions at wavevectors that do not exist on the coarse mesh and compare with the *standard* d_3 dynamic bubble of `cellconstructor.Spectral.` the third-order tensor route (here fed with the *exact analytic* $\Phi^{(3)}$ of the bond model, centered and ASR-projected, interpolated by `ForceTensor.Tensor3`), with the same k -integration mesh as the Lanczos fine mesh and the same smearing. The toy is purely cubic and weakly coupled, so the perturbative bubble and the (resummed) TDSCHA Lanczos should agree closely; the Lanczos runs with the D4 terms disabled to match the diagram content. Figure 8 shows the interpolated-Lanczos spectral functions on top of the bubble ones at interpolated wavevectors of the $(1, 1, 16)$ mesh from a coarse $(1, 1, 4)$ ensemble ($N_I = 8000$): peak positions and line shapes agree at the level of the energy-grid resolution ($\approx 2.3 \text{ cm}^{-1}$), on anharmonic shifts of $15 \dots 40 \text{ cm}^{-1}$:

q	band	ω_{SSCHA} (cm^{-1})	peak (Lanczos interp) (cm^{-1})	peak (d_3 bubble) (cm^{-1})
2/16	4	2205.3	2185.6	2183.2
2/16	5	2250.6	2214.2	2216.6
6/16	4	2013.4	1994.9	1992.5
6/16	5	2058.5	2040.2	2040.2

11 Fourth order: spectral functions where D_4 is fundamental

The bubble benchmark of Sec. 10 validates the *cubic* interpolation against an analytic diagram, with D_4 switched off. The four-phonon vertex is, however, interpolated on the same footing: its four-field Bloch averages are evaluated off-grid exactly like the D_3 ones and rescaled by $s_4 = N_c/N_f$ (Sec. 4). The only asymmetry is that the D_4 term rides a single plain-window pass rather than the designed multitaper passes used for D_3 (Sec. 6): its range is shorter and its N_c/N_f weight

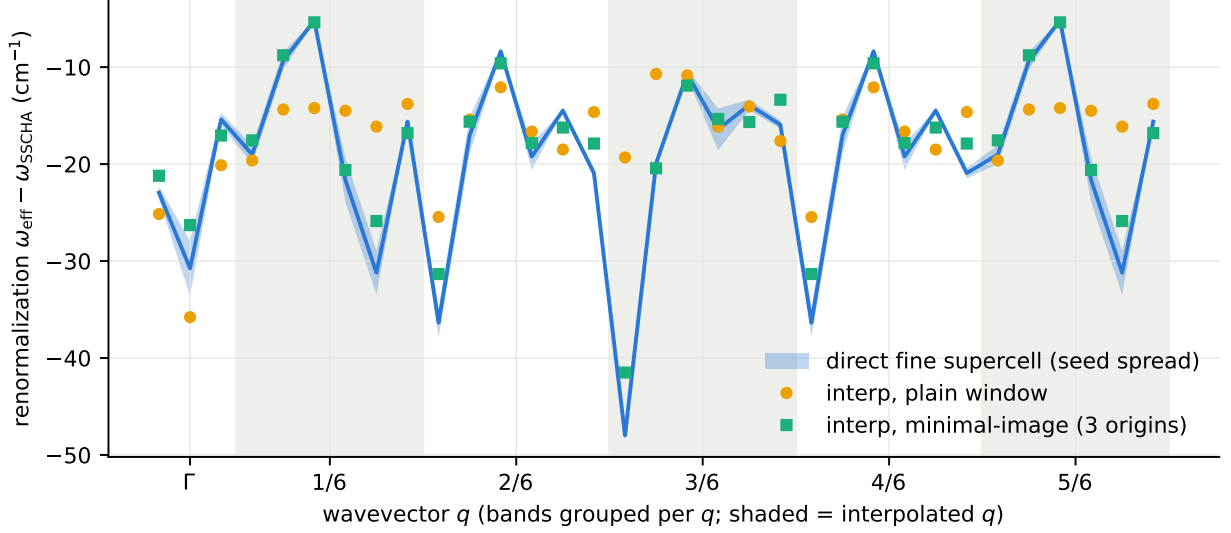


Figure 7: Mode-resolved anharmonic renormalization on the fine (1,1,6) mesh: direct fine-supercell reference (two independent seeds indicate the stochastic spread) versus the interpolated Lanczos from the coarse (1,1,3) ensemble with the plain window and with the designed minimal-image windows (3 origins). Vertical guides mark the interpolated (non-commensurate) wavevectors, including the non-TRI pair $q = \pm 1/6$.

smaller, so the extra windowed passes are not spent on it. This section validates that choice end-to-end.

There is no simple analytic diagram to compare against, because the static one-phonon self-energy of the *bare* quartic vertex — the tadpole $\frac{1}{2} \sum_k d_4(q\nu, -q\nu, k\mu, -k\mu) \chi_{k\mu}$ — is exactly the quantity SSCHA already resums into the auxiliary frequency ω_{SSCHA} . A *purely* quartic model therefore produces *zero* anharmonic renormalization of the one-phonon line: with $D_3 = 0$ the initial single-phonon state never couples to the two-phonon sector, the Lanczos chain terminates at first step, and $\omega_{\text{eff}} = \omega_{\text{SSCHA}}$ identically (we verified this numerically to machine zero). What survives of D_4 in the TDSCHA response is its action *inside* the two-phonon continuum that D_3 opens; it is genuinely a fourth-order effect, but only visible together with a nonzero third order.

Accordingly we validate D_4 by internal consistency in a mixed regime. The anharmonic diatomic chain is given a moderate cubic bond term ($g_3 = 0.15$) and a *strong* quartic one ($g_4 = 2.0$), so that the quartic vertex reshapes the spectrum by several tens of cm^{-1} . We compute the full $D_3 + D_4$ one-phonon spectral function $A(q, \omega) = -\frac{1}{\pi} \text{Im} G(q, \omega)$ (continued-fraction Green function, $N_{\text{steps}} = 110$, smearing $\approx 6.6 \text{ cm}^{-1}$) at wavevectors of the fine (1,1,9) mesh, in two ways: interpolated from a coarse (1,1,3) ensemble (minimal-image windows, two origins) and directly from an independent (1,1,9) ensemble of the *same* chain ($N_I = 6000$ each). As a control we also record each spectrum with D_4 switched off.

Figure 9 and the table below show the outcome at three interpolated wavevectors. The quartic vertex is fundamental: it hardens the peak by $30 \dots 56 \text{ cm}^{-1}$ relative to the D_3 -only line (dashed), a shift comparable to the SSCHA frequency’s own anharmonic budget and far larger than any interpolation error. The interpolated full spectrum (aqua) nonetheless lands on the direct fine-mesh reference (blue) to within one energy-grid step, $\lesssim 2 \text{ cm}^{-1}$ on the peak and $L_1 \approx 0.15 \dots 0.22$ on the whole normalized line shape (with $L_1 = \int |\hat{A}_{\text{interp}} - \hat{A}_{\text{direct}}| d\omega$, $\hat{A}$ normalized to unit area). Dropping D_4 from the interpolation raises L_1 to $\approx 1.7 \dots 1.8$ — an order of magnitude worse, the two line shapes being essentially disjoint. The fourth order thus interpolates correctly and is indispensable, even though it never sees the designed windows.

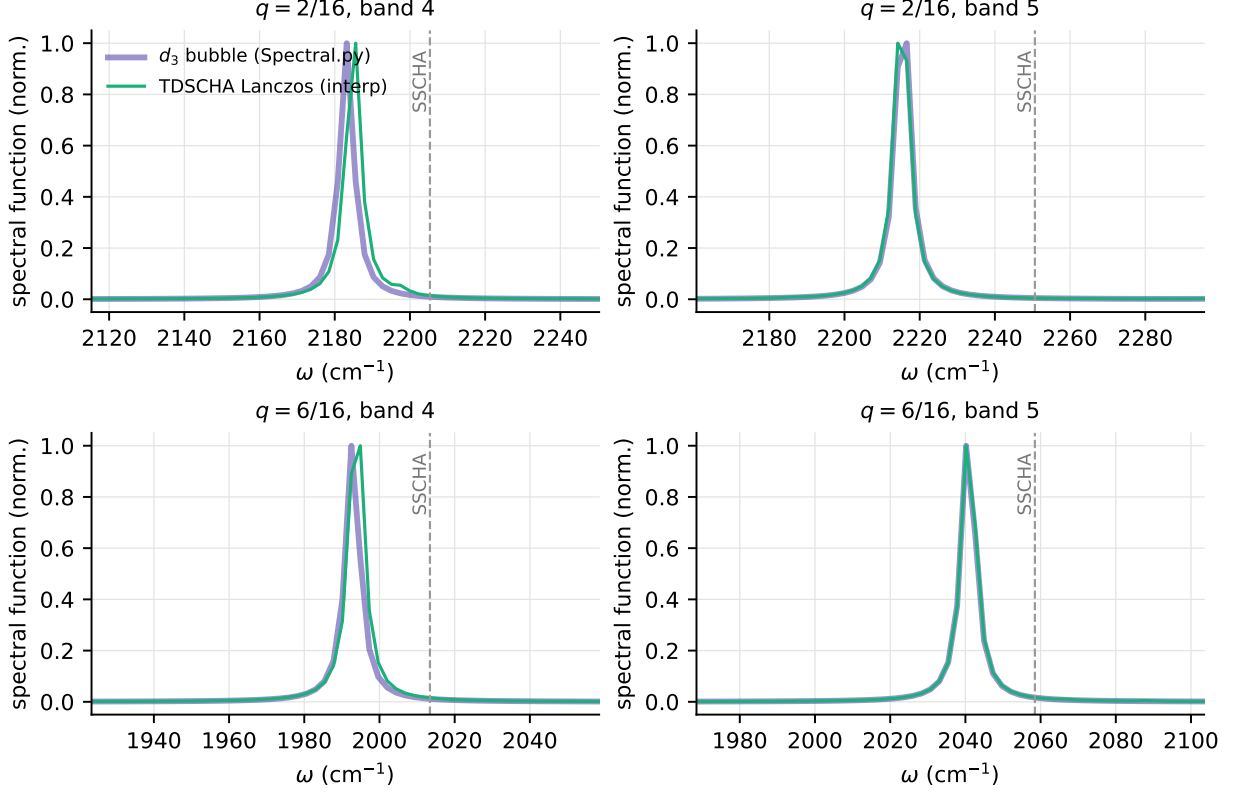


Figure 8: One-phonon spectral functions at interpolated wavevectors ($q = 2/16$ and $6/16$ along the chain, two optical bands): fine-mesh TDSCHA Lanczos interpolated from the coarse $(1,1,4)$ ensemble (minimal-image windows) versus the standard d_3 dynamic bubble of `cellconstructor.Spectral` with the exact centered $\Phi^{(3)}$ tensor on the same k -mesh and smearing. The SSCHA frequency is marked.

q	band	ω_{SSCHA} (cm^{-1})	peak (cm^{-1})		D_4 shift (cm^{-1})	interp-dir (cm^{-1})	L_1 full / D_3
			D_3	D_3+D_4			
1/9	5	2258.0	2198.8	2237.8	+39.0	+1.9	0.22/1.81
2/9	5	2177.4	2104.0	2159.8	+55.8	0.0	0.15/1.80
4/9	4	1951.4	1907.0	1936.7	+29.7	-1.9	0.17/1.70

Peaks are the direct fine-mesh values; the “ D_4 shift” is D_3+D_4 minus D_3 -only (direct), and “interp-dir” is the interpolated D_3+D_4 peak minus the direct one. At the operator level the two design choices are exact rather than merely accurate: the D_4 block of $\mathcal{L}$ scales strictly linearly in s_4 and is bit-identical across the plain, minimal-image and ASR window designs (it is produced by the same plain-window pass in all three), which the test suite checks directly.

12 Limitations and outlook

(i) Long-range (polar) parts of $\Phi^{(3)}$ are not separated analytically (the same limitation as the d_3 route of `Spectral.py`); LO-TO for the interpolated dynamical matrix is phase-2 work. (ii) The fine mesh adds no new microscopic information: noise at nearby fine wavevectors is correlated over the coarse-mesh spacing, and the genuine q -resolution of the *anharmonic* content is set by the coarse supercell — the physical gain is the dense two-phonon continuum and the interpolated harmonic dispersion. (iii) The ASR-exact design (Sec. 6) enforces only the translational sum rules, like `Apply_ASR`; rotational/Born-Huang invariances are not imposed, and the doubled-

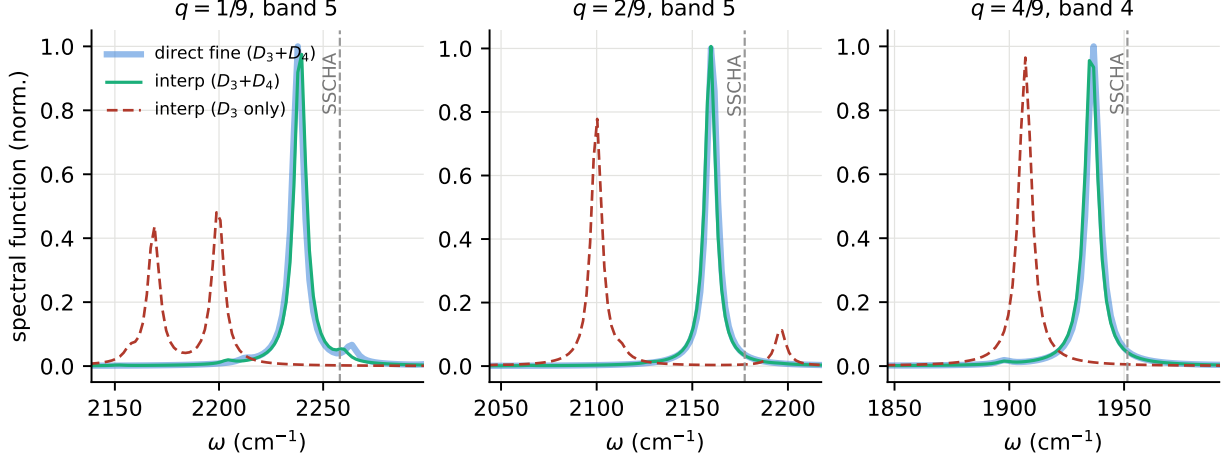


Figure 9: One-phonon spectral functions in a D_4 -dominated regime ($g_3 = 0.15$, $g_4 = 2.0$) at interpolated wavevectors of the $(1, 1, 9)$ mesh. Blue: direct fine-mesh TDSCHA Lanczos (D_3+D_4). Aqua: interpolated from the coarse $(1, 1, 3)$ ensemble (D_3+D_4) — it overlays the direct reference. Red dashed: the same interpolation with the fourth order dropped (D_3 only), displaced by the quartic hardening of $30 \dots 56 \text{ cm}^{-1}$. The SSCHA frequency is marked; curves normalized to the direct peak.

support fit realizes the projected target with a small residual (Sec. 6.3) that larger supports reduce systematically. (iv) The distributed (MPI config-sliced) loader and the IR/Raman $\sqrt{N_f}$ prefactor for the interpolated class remain to be wired. (v) Off-mesh perturbation wavevectors (spectral maps along arbitrary paths with little-group symmetrization) are a natural extension: all required machinery (NUDFT at arbitrary $\tilde{\mathbf{q}}$) is already in place.

A Derivation of the effective interpolation kernel

We derive Eq. (6) in one dimension (the three-dimensional case is the per-dimension product). Let $u(s)$, $s \in \mathbb{Z}$, be the supercell-periodic configuration, $u(s+L) = u(s)$, and let the windowed field at wavevector $\tilde{q}$ (in units where the phase is $e^{-2\pi i \tilde{q}s}$) be

$$\tilde{x}_W(\tilde{q}) = \frac{1}{\sqrt{L}} \sum_s W(s) e^{-2\pi i \tilde{q}s} u(s), \quad (21)$$

with W supported on one period ($W = 1$ on $[0, L)$ for the plain estimator). Consider the three-field product with windows (z, w, v) on the slots and exact conservation $\tilde{q}_1 + \tilde{q}_2 = \tilde{q}_p$:

$$P = \overline{\tilde{z}(\tilde{q}_p)} \tilde{x}_w(\tilde{q}_1) \tilde{y}_v(\tilde{q}_2) = \frac{1}{L^{3/2}} \sum_{s_1 s_2 s_3} z(s_3) w(s_1) v(s_2) e^{2\pi i (\tilde{q}_p s_3 - \tilde{q}_1 s_1 - \tilde{q}_2 s_2)} \bar{u}(s_3) u(s_1) \delta f(s_2). \quad (22)$$

Taking the ensemble average and using translation invariance, $\langle \bar{u}(s_3) u(s_1) \delta f(s_2) \rangle = c_3(s_1 - s_3, s_2 - s_3)$ with c_3 evaluated at the *actual* integer differences (the physical correlation of the periodic system, which itself is the image sum $c_3(\delta_1, \delta_2) = \sum_{n_1 n_2} c_3^\infty(\delta_1 + n_1 L, \delta_2 + n_2 L)$ of the bulk correlation). Substituting $\delta_i = s_i - s_3$ and using $\tilde{q}_p - \tilde{q}_1 - \tilde{q}_2 = 0$ *exactly*, the s_3 dependence of the phase cancels:

$$\langle P \rangle = \frac{1}{L^{3/2}} \sum_{\delta_1 \delta_2} \underbrace{\left[\sum_{s_3} z(s_3) w(s_3 + \delta_1) v(s_3 + \delta_2) \right]}_{S_{z,w,v}(\delta_1, \delta_2)} c_3(\delta_1, \delta_2) e^{-2\pi i (\tilde{q}_1 \delta_1 + \tilde{q}_2 \delta_2)}. \quad (23)$$

Because the windows are zero-padded (support $\leq L$), the differences δ_i range over $(-L, L)$ and each *physical* image of c_3^∞ is weighted by S at its actual (unwrapped) difference and carries its actual phase — this is where the centering choice lives. Two consequences:

- **Commensurate limit.** If $\tilde{q}_i$ are commensurate, all images of a difference class $(\delta_1, \delta_2) \bmod L$ share the same phase, so only the class sums $\sum_{\mathbf{n}} S(\delta + L\mathbf{n})$ matter. If they equal L (partition of unity), Eq. (23) collapses to the exact coarse average — for *any* window design satisfying the constraint. The plain window satisfies it identically ($S = L - \text{spread}$, class sums $= L$).
- **Off-grid.** The kernel S/L is the image-assignment rule of the interpolation. The minimal-image target of Sec. 3.2 makes it the indicator of the most compact image (ties split), reproducing zero-padded tensor centering.

The same algebra applies verbatim to every D3 term (the y field can sit in any slot) and, with one more field, to the D4 terms, whose kernel is the window quadruple correlation. Note also that no origin appears in Eq. (23): translating the window (equivalently, rolling the data) leaves the expectation invariant, so origin averaging only reduces variance.

B Gaussian integration by parts and the pre-filter identity

For the Gaussian density ρ_Υ with covariance $\langle u_a u_b \rangle = \Upsilon_{ab}^{-1}$, Wick/Isserlis integration by parts gives, for any observable $F(u)$, $\langle u_a F \rangle = \Upsilon_{ab}^{-1} \langle \partial F / \partial u_b \rangle$. Applying it twice to the D3 average (with $\delta f = f - f_{\text{SSCHA}} - \langle f \rangle$, whose second derivative equals that of the full force $f = -\partial V / \partial R$):

$$\langle u_a u_b \delta f_c \rangle = \underbrace{\Upsilon_{ab}^{-1} \langle \delta f_c \rangle}_{=0} + \Upsilon_{aa'}^{-1} \Upsilon_{bb'}^{-1} \left\langle \frac{\partial^2 \delta f_c}{\partial u_{a'} \partial u_{b'}} \right\rangle = -\Upsilon_{aa'}^{-1} \Upsilon_{bb'}^{-1} \langle \partial_{a'b'c}^3 V \rangle. \quad (24)$$

In the SSCHA mode basis Υ^{-1} is diagonal with eigenvalues $f_\psi(\mathbf{q}, \nu) = (1 + 2n_{\mathbf{q}\nu})/2\omega_{\mathbf{q}\nu}$: every *displacement* leg of the correlation carries a propagator factor f_ψ , while force legs carry none. Since the Lanczos kernel multiplies every displacement leg by $f_\Upsilon = 1/f_\psi$ (the $r_1 = f_\Upsilon x$ fields), the combination it averages is the *bare* vertex contraction. The pre-filter of Sec. 5 simply moves the f_Υ multiplication *before* the off-grid transform: on the coarse grid $f_\Upsilon x$ is computed exactly (mode-diagonal), and its real-space transform is a configuration whose correlations with y are contractions of the short-ranged $\Phi^{(3)}$ (in the Cartesian basis, before any polarization projection — the projections are applied at the fine wavevectors analytically). Off-grid, interpolating the filtered fields therefore interpolates a Φ -ranged object; the exact fine-side factors $f_\psi(\tilde{\mathbf{q}})$ that the kernel formulas still require are supplied analytically (folded into the $\alpha^{(1)}$ blocks and set to unity in the kernel tables).

For the four-field averages the same expansion produces, besides the connected $\langle \partial^4 V \rangle$ term, disconnected pieces proportional to $\langle u \otimes \delta f \rangle = \Upsilon^{-1} (\langle \partial^2 V \rangle - \Phi)$, which vanish when Φ is the converged SSCHA matrix. This is the stationarity condition under which the uniform vertex rescaling of Sec. 4 is exact.

C Reproducibility

All numbers and figures in this report are produced by the scripts in `tests/test_interpolation/` (toy model and physics tests) and the benchmark/figure scripts referenced in the repository (`report/interpolation/scripts/`, including `asr_data.py` and `asr_figs.py` for Sec. 6); the decisive physics comparisons are enforced as regression tests (`test_physics_interp.py`, `test_windows.py`, `test_asr_windows.py`). The atom-resolved geometry checks are in `test_atomic_windows.py`. The SnTe oracle and response numbers of Sec. 7 are tracked in `SnTe_FF/Spectral/TDSCHA_Interpolate/issues`; the relevant scripts/data are `kernel_study.py`, `atomic_d3_N300_s20.dat` and `atomic_d3_N1000_s30.dat`.

References

- [1] L. Monacelli and F. Mauri, *Time-dependent self-consistent harmonic approximation*, Phys. Rev. B **103**, 104305 (2021).
- [2] TDSCCHA q -space Lanczos implementation, `tdscha.QSpaceLanczos` (this repository).
- [3] R. Bianco et al., Phys. Rev. B **97**, 214101 (2018) (dynamic bubble conventions used by `cellconstructor.Spectral`).